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Faculty of Computing Department of Software Engineering
Operating system and system programming

Title: linux lite os
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Introduction to Linux Lite OS

In the vast ecosystem of Linux distributions, where options range from the technically complex Arch Linux to the enterprise-focused Red Hat, Linux Lite carves out a unique and vital niche. Launched on October 26, 2012, by New Zealand-based developer Jerry Bezencon, Linux Lite is a free, open-source operating system designed with a singular, user-centric mission: to act as a "gateway" into the world of Linux .

Ubuntu LTS (Long Term Support) editions serve as the foundation for Linux Lite, a free and thin linux system. To make Linux accessible to customers who were used to Windows but were searching for a free and open-source substitute. One of linux lite's main goals was to develop a system that would function well on outdated or low-spec hardware so that users may prolong the life of their current Computers.

The first Linux Lite version which was based on Ubuntu 12.04 LTS included the Xfce desktop environment, which was selected for its ease of use and low resource use. Linux Lite has remained based on Ubuntu's LTS releases over the years, guaranteeing stability, security upgrades, and longterm maintenance. Pre-installed software packages that are often used by most users, including a web browser, email client, media player, office suite, and system tools, are included with every edition of Linux Lite. With this method, new users can get a completely functional operating system right out of the box.

The background of Linux Lite is rooted in a specific moment in computing history. As Microsoft ended support for the long-beloved Windows XP in April 2014, millions of perfectly functional personal computers faced an unprecedented security risk, destined to become obsolete or be replaced by expensive new hardware running modern, resource-hungry operating systems . Recognizing this "vintage PC" crisis, as well as the growing complexity of standard Linux desktops like Ubuntu’s Unity, the creators of Linux Lite identified a clear opportunity: to provide a bridge for frustrated Windows users .

The motivation behind the project is twofold and pragmatic. First and foremost, Linux Lite aims to dispel the myth that Linux is difficult to use . Secondly, it seeks to extend the lifespan of aging hardware . By offering a streamlined, familiar interface that mirrors the layout of Windows, Linux Lite reduces the learning curve dramatically. It strips away the bloated code and resource-heavy animations found in mainstream operating systems, providing a "Simple, Fast, Free" experience that runs efficiently on computers with as little as 1GB of RAM . It is not just an operating system; it is a philosophy that high-performance

computing should be accessible to everyone, regardless of their technical background or the age of their machine.

Objectives

- To install Linux Lite 7.8 in Oracle VirtualBox successfully.
- To learn the system-level programming capabilities using Linux Lite.
- To implement and understand the pivot-root() system call.
- To explore Linux filesystem support and virtualization technology.
- To show the lightweight nature of Linux lite.

Requirements

Linux lite is probably the best of all lightweight Linux distributions. Based on Ubuntu LTS and designed with user-friendliness in mind, Linux Lite uses the Xfce desktop to create a UI that's clean and simple but doesn't miss out on features.

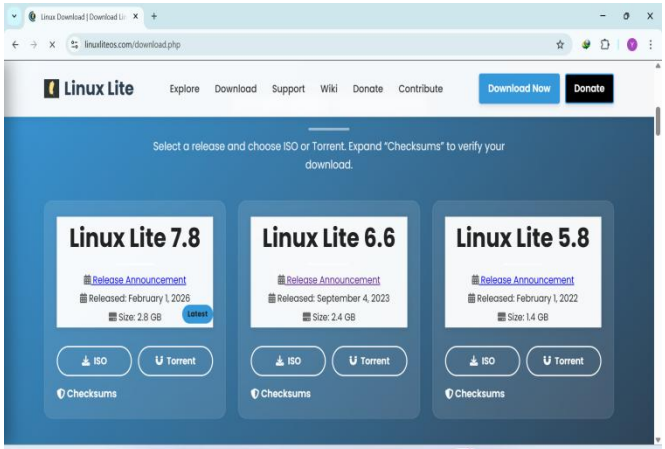
I. Hardware requirements
recommended requirements:
• CPU: Dual core Intel or AMD x64 processor with 2 GHz or higher.
• RAM: 2 - 4 GB is recommended for a more comfortable experience.
• Storage: 10 GB of disk space.
• Graphics: VGA screen with 1024x768 resolution.
minimum requirements:
• CPU: 1 GHz processor.
• RAM: 1 GB.
• Storage: 8 GB of drive space.

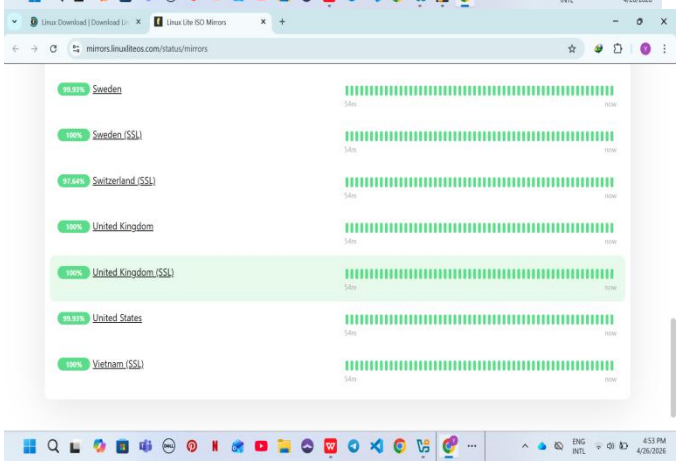
Linux lite is created for the sole purpose of repurposing older computers to make them compatible with modern technology so its installation requirements are very limited. But installing it in a virtual machine would require a lot more specifications since virtualization can be demanding.

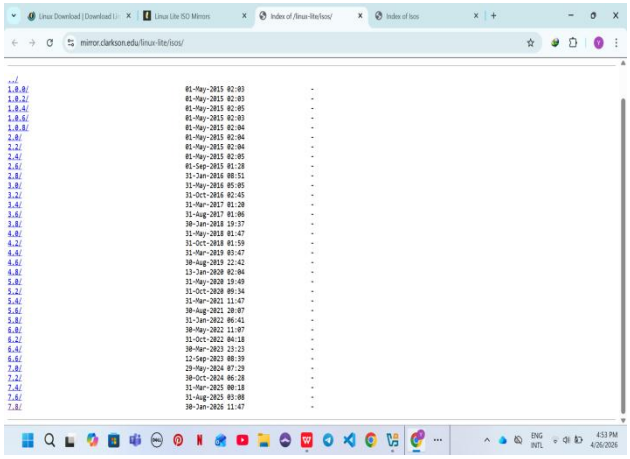
II. Software requirements
- Linux Lite 7.0 ISO file which can be downloaded from their official website
If you want to install it in Oracle VirtualBox.
- Oracle VirtualBox
- Host OS (e.g. Windows/Linux).
- It will need at least 1GB for Linux and 2-4GB for the virtual machine
- It requires a modern multi-core processor like intel i5 or AMD RYZEN

Installation steps

✧ Download Linux Lite 7.8 ISO from the official website.





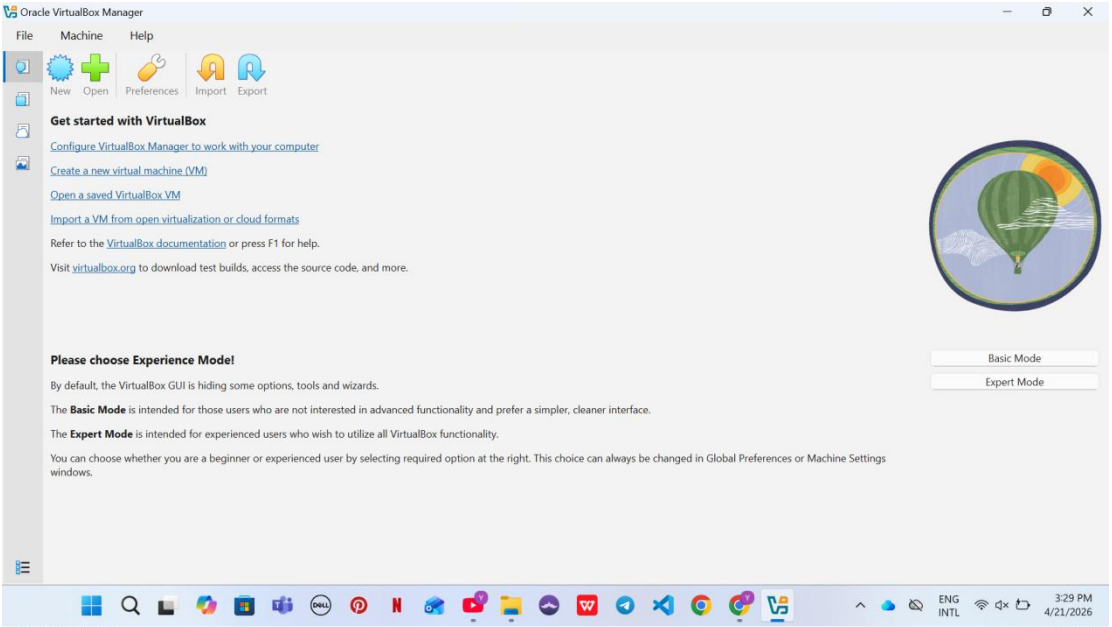


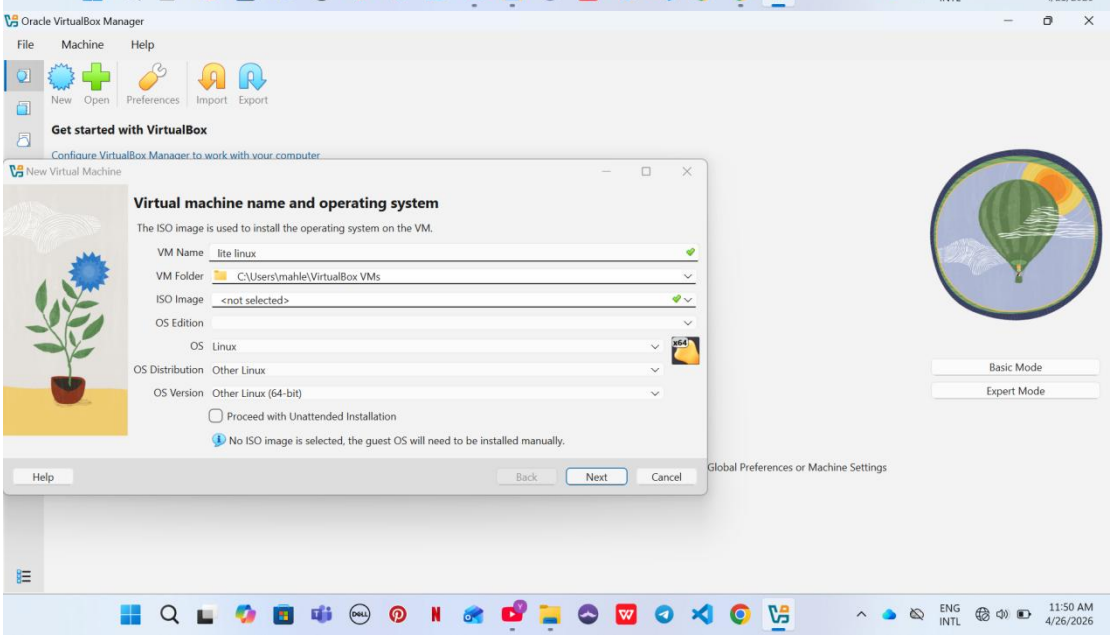
✧ Install VirtualBox

-Go to https://www.virtualbox.org

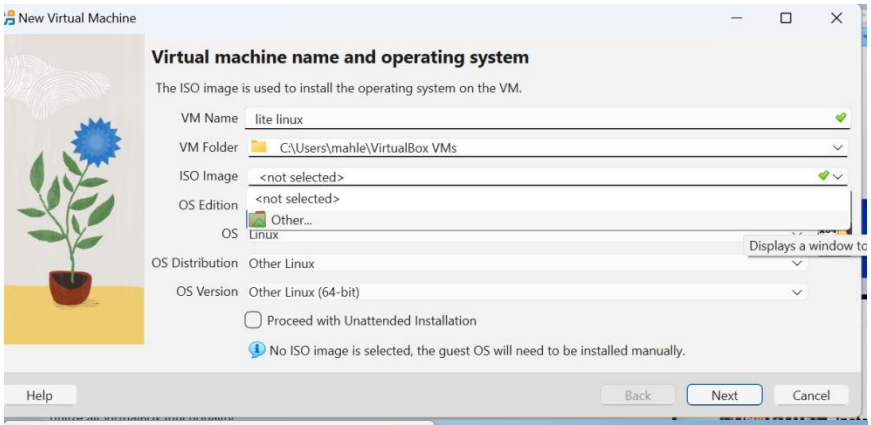
- Download and install the version for your OS (Windows/macOS/Linux)

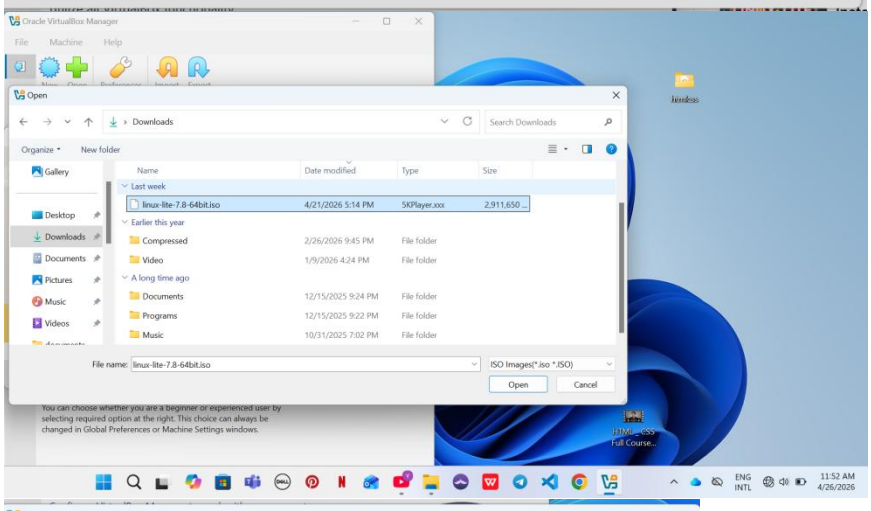
✧ Open Oracle VirtualBox and create a new virtual machine.

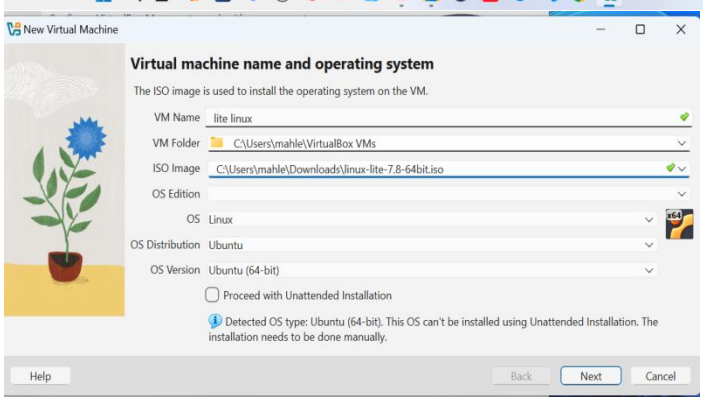




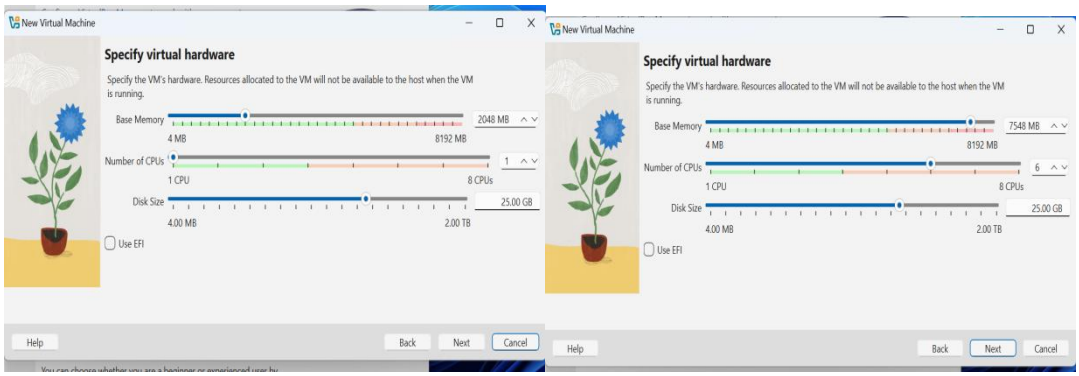
✧ Link the iso file you downloaded to the virtual machine under “ OS Edition’’



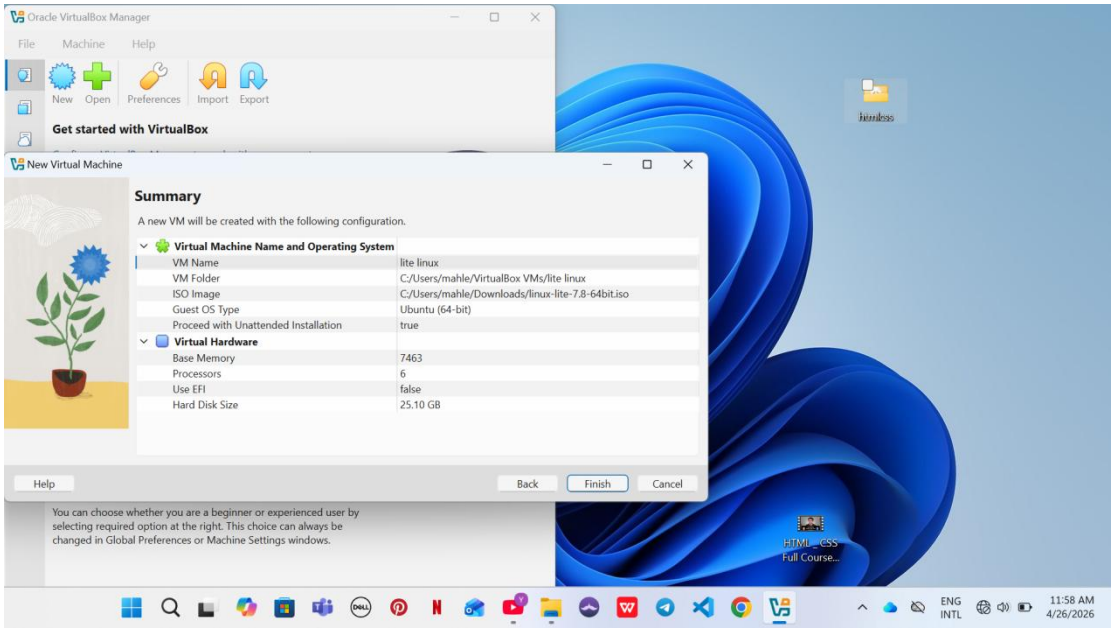




✧ Assign memory and create a virtual hard disk (Minimum **2048 MB ram** and at least 30GB).



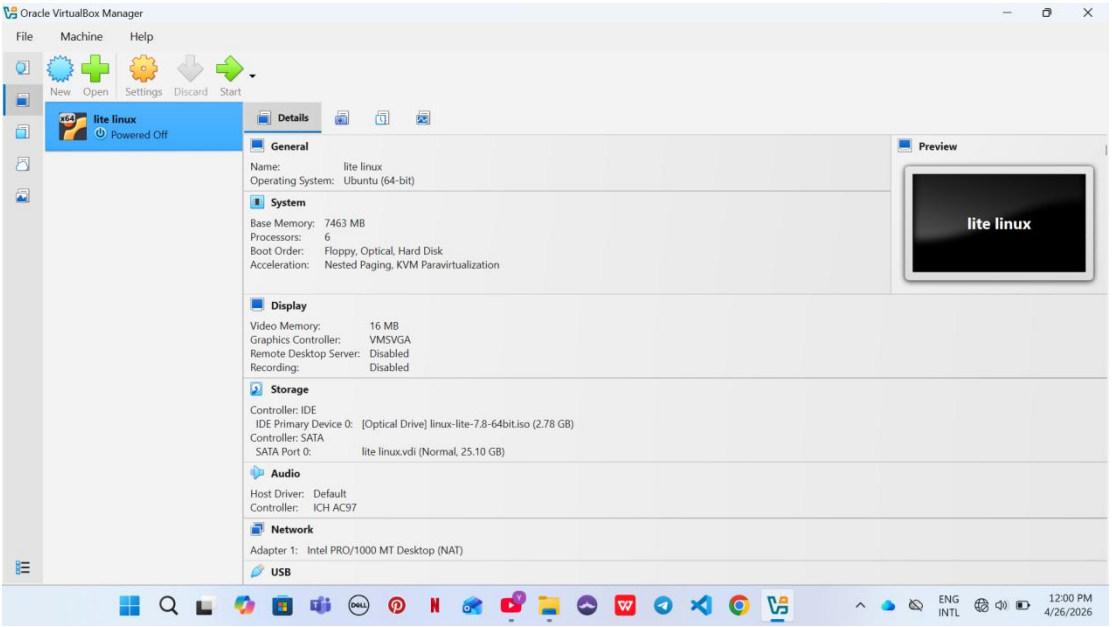
✧ After allocating memory click the ''finish'' button

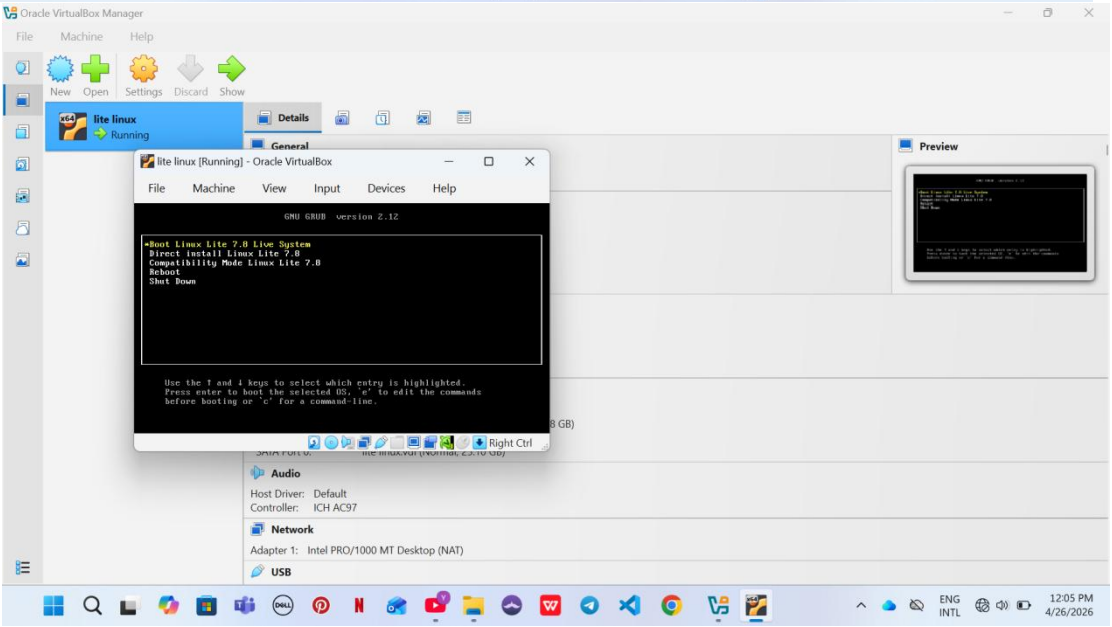


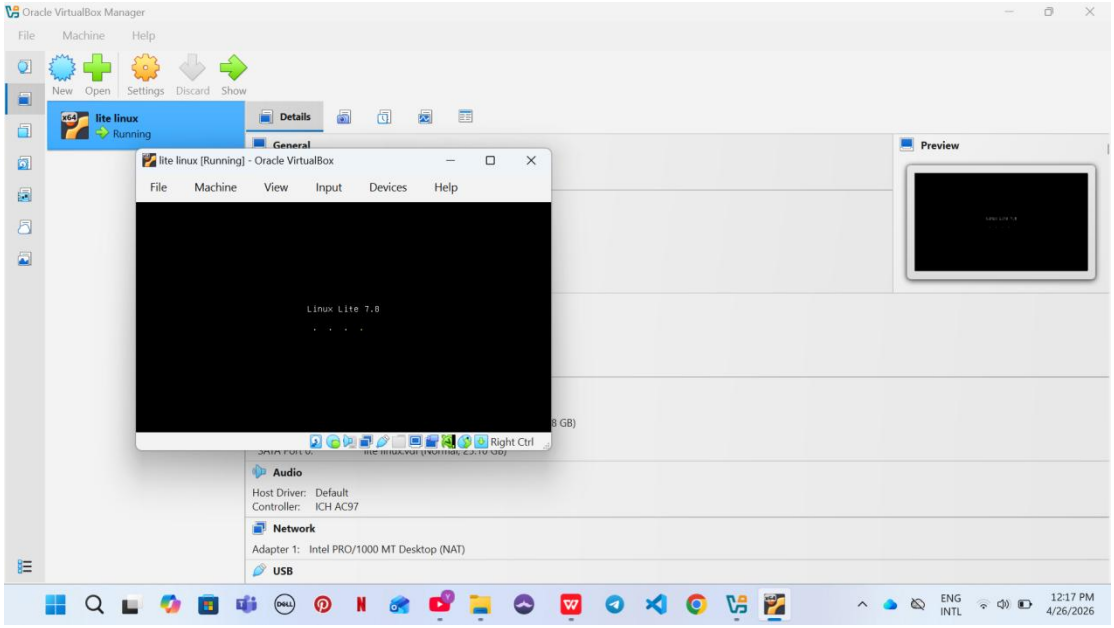
✧ Attach Linux Lite ISO as the boot medium and start the VM and follow the on-screen instructions to install Linux Lite.

1. Click **Start**
2. Boot into the **Live Linux Lite** session.

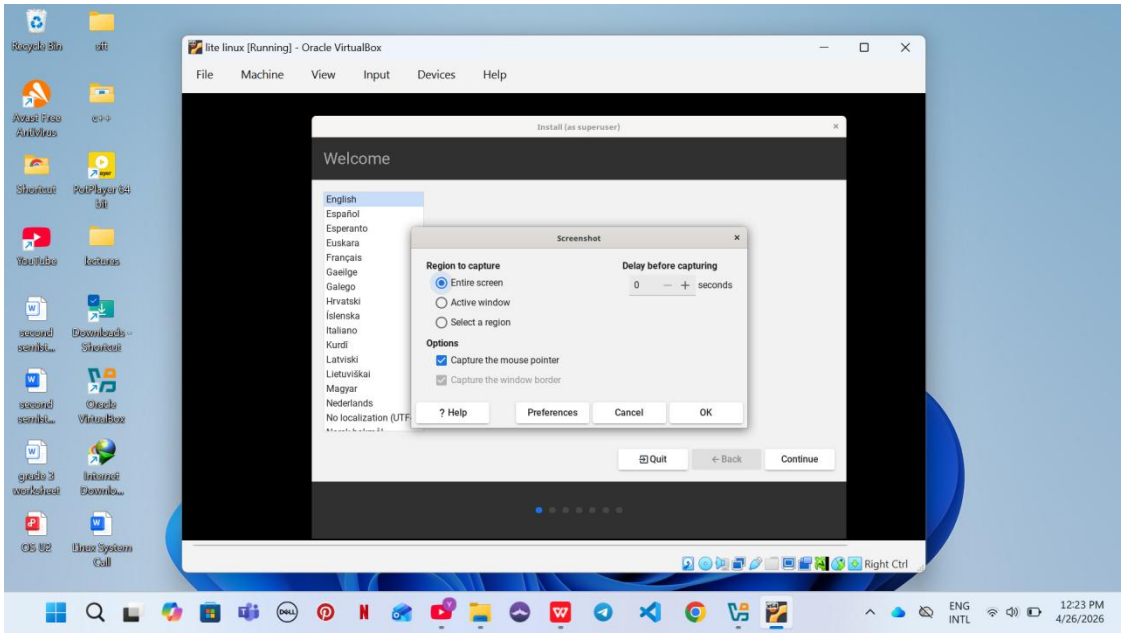
-This step will start the actual installation of linux lite os



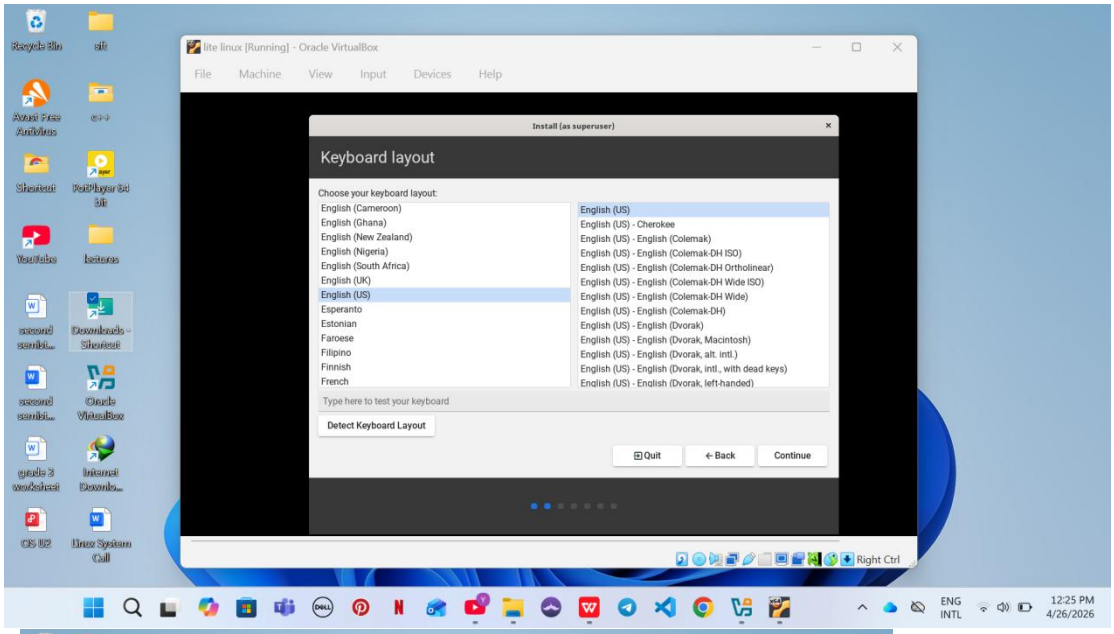


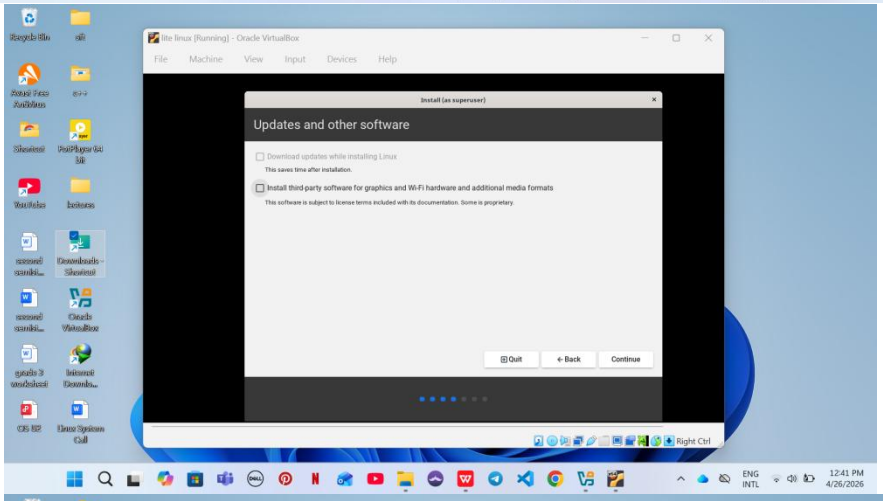


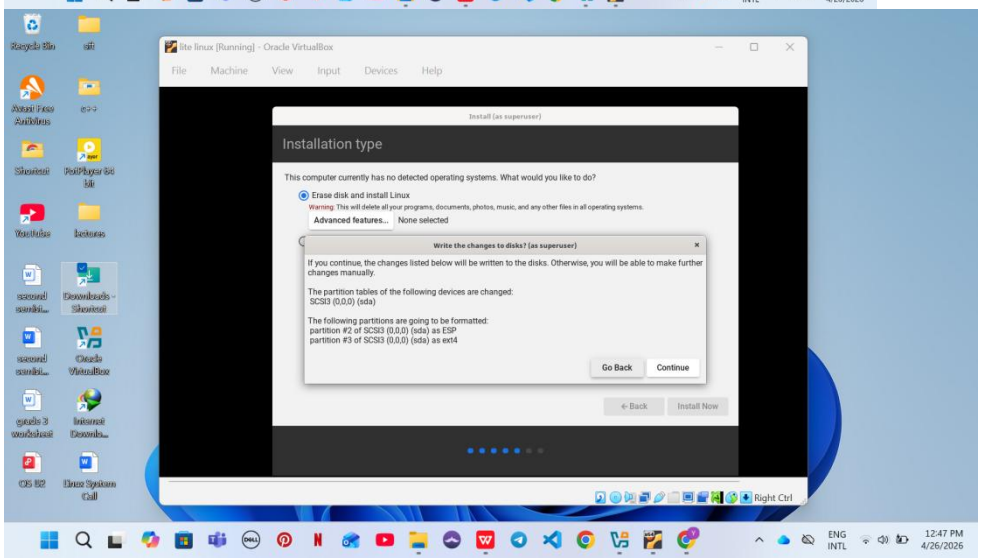
✧ Select language



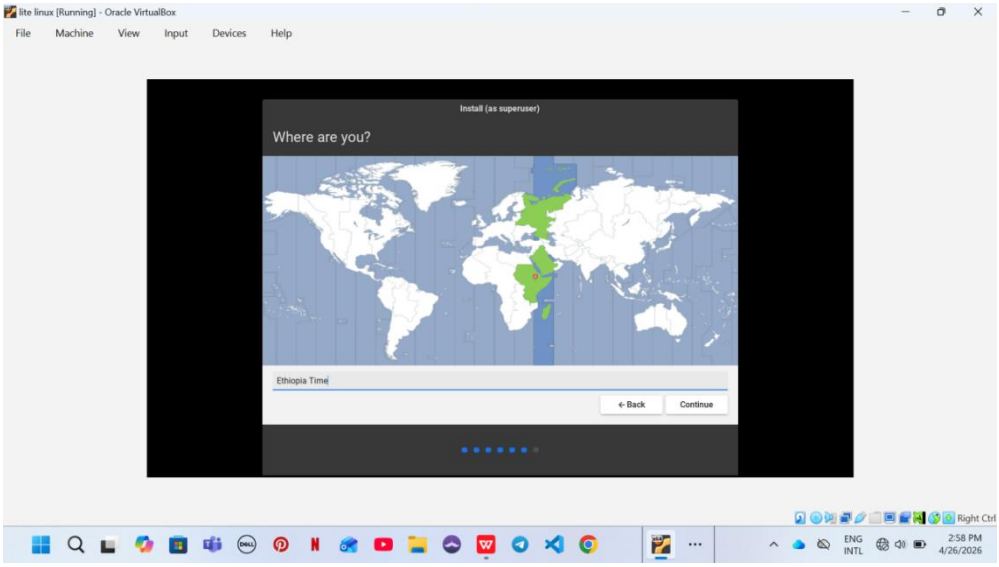
✧ Select keyboard layout then click “continue”



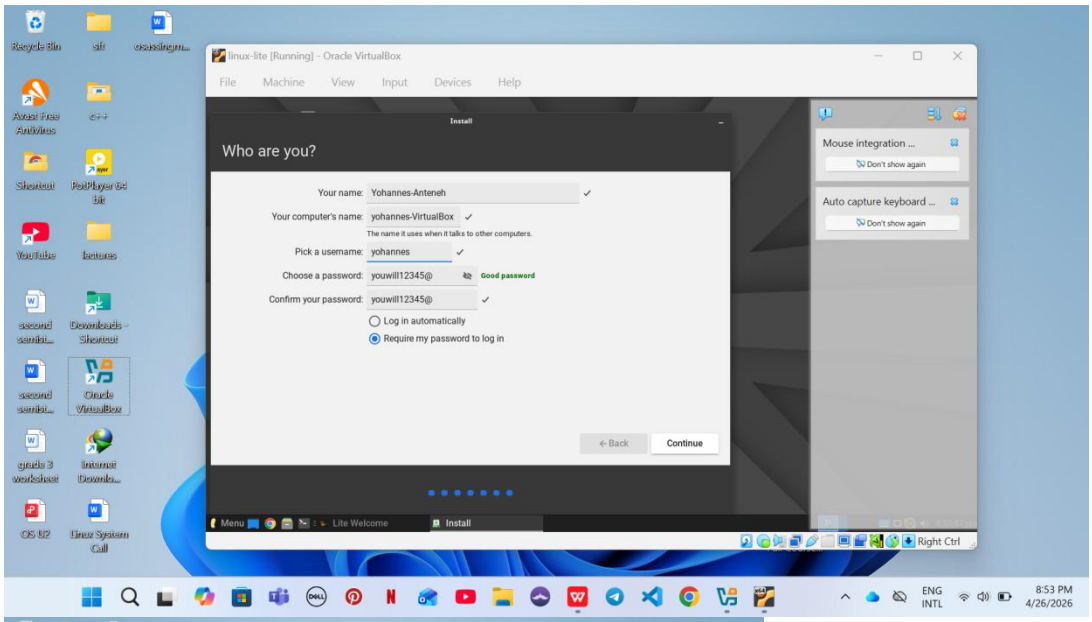


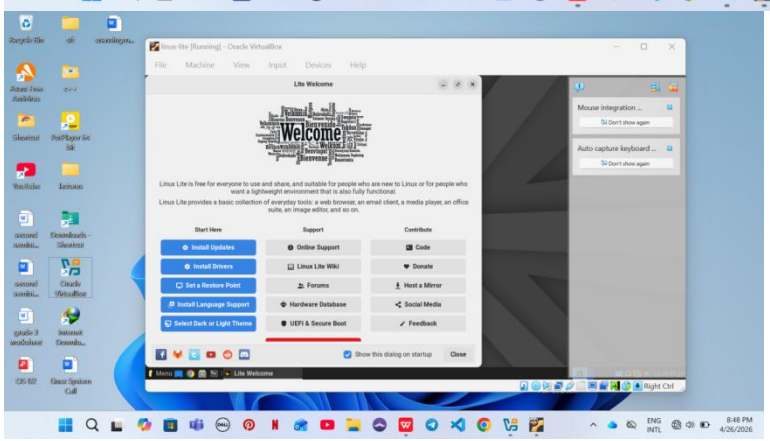


✧ Set time zone

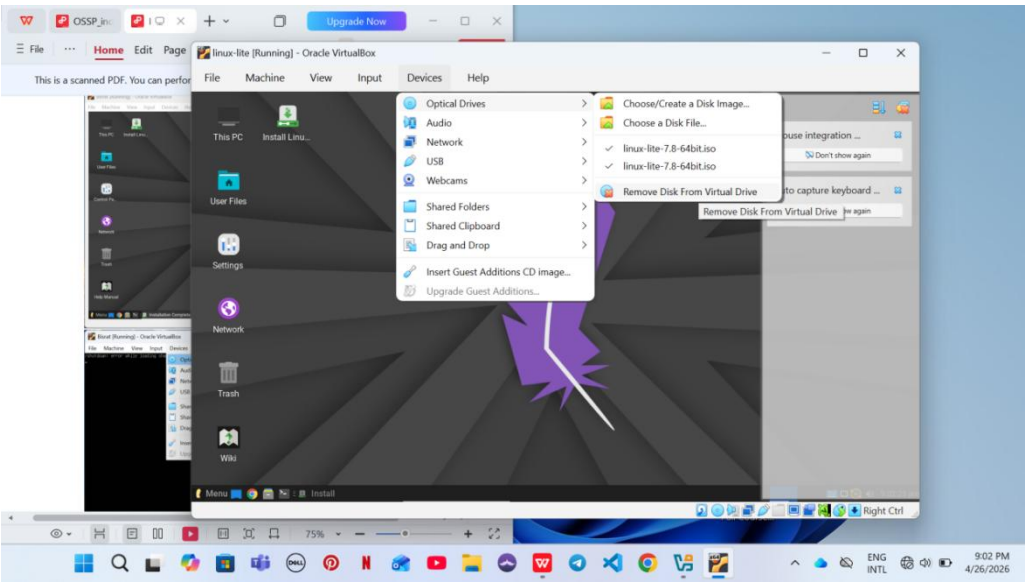


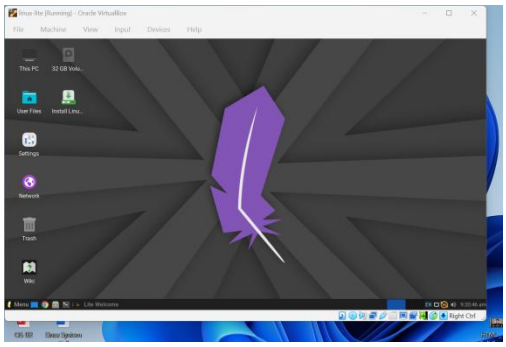
✧ Set new username password

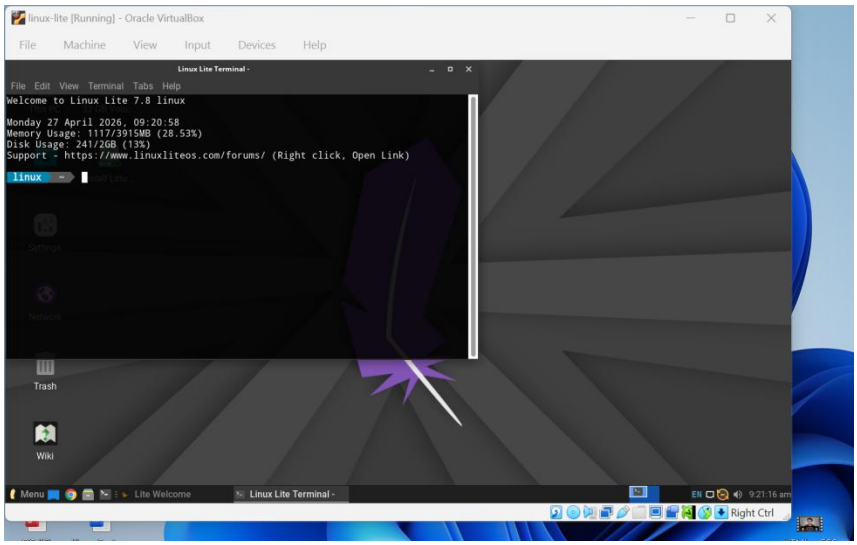




✧ Remove ISO After Install







Issue faced

-Linux Lite runs very slowly.
-Installer hangs or crashes midway.
-Virtual disk too small.

Solution

-Allocate at least 20 GB when creating VM and If already created Use VirtualBox tools to resize disk, Then extend partition inside Linux
-Increase resources:
RAM: at least 2 GB (preferably 4 GB)
CPU: 2 cores or more

File system support

Linux Lite 7.0 mainly uses the ext4 file system. This choice is made based on several advantages that lies with the system's focus on simplicity, reliability, and user-friendliness.
Being based on Ubuntu 22.04, Linux Lite 7.0 inherits strong support for a variety of additional filesystems, allowing users to easily interact with external drives, dual-boot setups, and legacy systems.

- **FAT32**

- Compatibility with flash drives, memory cards, and older Windows systems.
- Full read/write.
- Limited to 4GB per file and 8TB per partition. Commonly used for bootable USB media and simple file transfers.

- **NTFS**

- Used by Windows systems for internal and external storage.
- Full read/write via the ntfs-3g driver (preinstalled in most setups).
- Allows seamless access to Windows drives for file sharing in dual-boot systems.

- **exFAT**

- Preferred for large-capacity flash storage (SDXC cards, USB drives).
- Full native read/write support since Linux kernel 5.4.
- No file size limit (unlike FAT32). Ideal for large file transfers between Linux, Windows, and macOS.

- **Btrfs**

- Advanced Linux users requiring features like snapshots, checksums, and built-in RAID
- Available via kernel and btrfs-progs package.

- Not used by default due to higher complexity and suitability for enterprise-level features.

- **ZFS**

- Enterprise-class filesystem with a focus on data protection, storage pooling, and scalability.
- Optional. Can be enabled by installing the zfsutils-linux package.
- Not installed by default due to licensing conflicts with the Linux kernel (ZFS uses CDDL, Linux uses GPL).

- **HFS+**

- Older macOS systems.
- Read supported by default; write support is possible but unreliable unless journaling is disabled.
- Safe for reading macOS drives but not recommended for writing.

- **APFS**

- Current macOS systems (High Sierra and later).
- Only basic read-only access via third-party tools like apfs-fuse.
- No official Linux kernel support or reliable write functionality due to Apple's proprietary implementation.

Advantages and Disadvantages

Advantages of Linux Lite:

Lightweight and Resource-Friendly: Linux Lite is designed to be lightweight, requiring minimal system resources to run smoothly, even on older hardware.
Easy to Use: It boasts a user-friendly interface and is considered beginner-friendly, particularly for those transitioning from Windows.
Cost-Effective: Linux Lite is free to use and doesn't require any license fees, making it a cost-effective alternative.
Pre-installed Essential Softwarem: Comes with useful apps like a web browser, office tools, and media players—ready to use immediately.
Free space and Open Source: As an open-source operating system, Linux Lite benefits from a large community of developers and users who contribute to its development and support.
Stable and Secure: Linux Lite is known for its stability and robust security features, making it a reliable choice for everyday use.
Reviving Old Hardware: Linux Lite can breathe new life into older computers and laptops, extending their lifespan and usefulness.

Disadvantages of Linux Lite:

Limited advanced features: Compared to more advanced distributions, it may lack cutting-edge tools and customization options.

Smaller Community than Major Distros: While Ubuntu has a massive community, Linux Lite's own community is smaller, so support may sometimes be limited.

Not Ideal for Heavy Tasks: Not designed for high-end gaming, video editing, or intensive computing workloads.

Software Compatibility Issues: Some Windows-specific programs won't run natively (you may need tools like Wine, which aren't always perfect).

Slower Updates Compared to Rolling Releases: It follows Ubuntu's release cycle, so it may not always have the newest software versions.

Limited Default Customization: Out of the box, it looks simple, but customization options are not as extensive as some other Linux distros.

Conclusion

The installation and usage of Linux Lite provide a user-friendly experience for both new and experienced users, offering a stable and fast operating system for everyday computing needs. By supporting a wide range of filesystems, including ext4, NTFS, FAT32, exfat, and more, Linux Lite ensures compatibility across various devices and platforms. Its ability to work with popular filesystems, combined with advanced features like Btrfs and ZFS, makes it a solid choice for those seeking a robust system for personal or professional use. Additionally, the Linux Lite community, with its active support forum and continuous updates, ensures that the system is always improving. By maintaining a focus on stability and ease of use, Linux Lite 1617 stands as an excellent choice for users transitioning from other operating systems, especially those coming from Windows.

Future Outlook / Recommendations

Enhanced Filesystem Support and Integration: While Linux Lite currently supports a wide range of filesystems, future updates could improve the integration of ZFS and Btrfs for broader usage, particularly for users requiring advanced storage features such as snapshots, data integrity, and compression. Fully integrating APFS support for macos users would also be a valuable enhancement.

Improved Software Ecosystem and Compatibility: Even though Linux Lite provides a solid software repository, expanding compatibility with more proprietary applications (especially for design, gaming, and productivity) would make it even more appealing to a broader audience. Additionally, better compatibility with Wine and Proton could allow more Windows applications to run seamlessly on Linux Lite.

User Experience Enhancements: To improve the user experience, further refinement of the graphical user interface (GUI) and system customization options could be beneficial. Providing an even more streamlined and accessible interface for beginners, along with advanced tools for power users, will ensure that Linux Lite remains a versatile and attractive choice for a wide range of users.

Community Growth and Documentation: The expansion of the Linux Lite community and user resources (such as more detailed documentation, tutorials, and video guides) could further enhance the support system for both new and experienced users.

Final Recommendations: For anyone looking to transition to Linux, especially those familiar with Windows, Linux Lite remains one of the best choices due to its ease of use, efficient design, and solid performance. As the system18 continues to evolve, embracing the above-mentioned recommendations would help solidify its position as a leading lightweight Linux distribution suitable for a wide range of users, from home desktops to educational and business environments.

Resource Efficiency: Operating system based virtualization allows for greater resource efficiency as containers do not need to emulate a complete hardware environment, which reduces resource overhead.

High Scalability: Containers can be quickly and easily scaled up or down depending on the demand, which makes it easy to respond to changes in the workload.

Easy Management: Containers are easy to manage as they can be managed through simple commands, which makes it easy to deploy and maintain large numbers of containers.

Reduced Costs: Operating system based virtualization can significantly reduce costs, as it requires fewer resources and infrastructure than traditional virtual machines.

Faster Deployment: Containers can be deployed quickly, reducing the time required to launch new applications or update existing ones.

Portability: Containers are highly portable, making it easy to move them from one environment to another without requiring changes to the underlying application.

Testing and Development: Virtualization lets users test other operating systems, software, or configurations in isolated environments without impacting the host system. This is particularly useful for developers and testers

Future development of Linux Lite will probably continue to refine its graphical user interface, expand hardware support and add modernized user features without compromising system performance. With new versions of the linux kernel and ubuntu LTS arriving in the next couple of years, linux lite should be in a good position to take advantage of the future improvements in system security, file system management and virtualization. It's open source community development also guarantees that feedback from users remains at the center of determining it's usability and pertinence.

Virtualization in Modern Operating Systems

Operating System-based Virtualization is also known as Containerization. It is a technology that allows multiple isolated user-space instances called containers to run on a single operating system (OS) kernel. Unlike traditional virtualization, where each virtual machine (VM) requires its own OS, OS-based virtualization allows the sharing of the same OS while providing separate environments for running applications.

Uses of Virtualization

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- The host OS kernel is shared among all containers, unlike full virtualization (e.g., VMs) where each VM has its own kernel.
- The kernel enforces isolation between containers using namespaces (for process, network, filesystem isolation) and cgroups (control groups) for resource allocation (CPU, memory, disk I/O, network).
- cgroups limit and prioritize resource usage (CPU, memory, disk, network) per container.
- The kernel ensures that a container cannot exceed its allocated resources (unless explicitly allowed).
- The overhead comes from kernel-level isolation mechanisms (namespaces, cgroups), but it's minimal compared to full virtualization.

Virtualization in Linux Lite 7.8

Linux Lite as a Guest Operating System (Running under a Virtual Machine)
Linux Lite 7.8 is suitable as a guest operating system in virtualization software such as Oracle VirtualBox, VMware, and QEMU/KVM.
In this project, I have installed Linux lite inside Oracle VirtualBox to create a secure, isolated environment in which to test and learn about system-level commands like mkdir(), and in which to test filesystem support. Oracle VirtualBox simulates hardware resources such as the CPU, RAM, and storage, and Linux Lite acts as if they were actual hardware. Due to its lightness and optimization, Linux Lite performs well even with minimal virtual resources. VirtualBox Guest Additions enhance the virtual experience further by enabling features such as:

- Seamless mouse integration
- Dynamic screen resizing
- Shared clipboard between host and guest
- Shared folders for easy file exchange